

Pablo Barttfeld

Independent Researcher, CONICET
Institute of Psychological Research
National University of Córdoba, Argentina

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RESEARCH PROFILE Cognitive neuroscientist working at the intersection of consciousness, metacognition and brain dynamics. My research combines neuroimaging (fMRI, EEG), computational modelling and behavioural paradigms to study how the brain sustains consciousness and decision-making under uncertainty. I lead the [Cognitive Sciences Group](#) at CONICET-UNC.

CURRENT POSITION **Independent Researcher, CONICET** 2016–present
Institute of Psychological Research · National University of Córdoba, Argentina

EDUCATION **Postgraduate Diploma in Data Science and Machine Learning** 2024
National University of Córdoba, Argentina

Postdoctoral Researcher 2013–2015
NeuroSpin / CEA-Saclay, France
Advisors: Stanislas Dehaene & Ghislaine Dehaene-Lambertz

Postdoctoral Researcher 2011–2013
Integrative Neuroscience Laboratory, Physics Department, University of Buenos Aires, Argentina
Advisor: Mariano Sigman

Ph.D. in Engineering 2004–2010
University of Buenos Aires (UBA), Argentina
Advisor: Mariano Sigman

Licenciatura (B.Sc.) in Biology 1997–2002
National University of Córdoba (UNC), Argentina

PUBLICATIONS 49 publications | 20 as first or last author | $h = 23$, 4,100 total citations.
* equal contribution.

Selected Publications

1. **Barttfeld P***, Uhrig L*, Sitt J*, Sigman M, Jarraya B, Dehaene S. Signature of consciousness in the dynamics of resting-state brain activity. *Proceedings of the National Academy of Sciences*. 2015.
2. **Barttfeld P**, Wicker B, McAleer P, Belin P, Cojan Y, Graziano M, Leiguarda R, Sigman M. Distinct patterns of functional brain connectivity correlate with objective performance and subjective beliefs. *Proceedings of the National Academy of Sciences*. 2013.
3. Comay N, Solovey G*, **Barttfeld P***. Confidence phenotypes: A unified computational account of value and decision certainty in reinforcement learning. *eLife*. 2026.
4. Della Bella G, Vélez Picatto A, Galván Rial S, Cukier S, Foa Torres G, Catanzariti M, Mateos D, Lamberti P, Cardeña E*, **Barttfeld P***. The Neurophenomenology of a Self-Induced Transcendental Visionary State: A Case Study. *NeuroImage*. 2026.
5. **Barttfeld P**, Comay N, Embon I, Solovey G. Pre- and post-decision signals of certainty in changing minds. *Trends in Neurosciences*. 2025.
6. Comay N, Della Bella G, Lamberti W, Sigman M, Solovey G*, **Barttfeld P***. The presence of irrelevant alternatives paradoxically increases confidence in perceptual decisions. *Cognition*. 2023.

7. Demertzi A, Tagliazucchi E, Dehaene S, Deco G, **Barttfeld P**, Raimondo F, Rohaut B, Schiff N, Owen A, Laureys S, Naccache L, Sitt J. Human consciousness is supported by dynamic complex patterns of brain signal coordination. *Science Advances*. 2019.
8. Della Bella G*, Zang D*, Gui P*, Mateos D, Sitt J, Bekinschtein T, *et al.*, **Barttfeld P**. Detection of EEG dynamic complex patterns in disorders of consciousness. *Communications Biology*. 2025.
9. **Barttfeld P**, Wicker B, Cukier S, Navarta S, Lew S, Sigman M. A big world network in ASD: Dynamical connectivity analysis reflects a deficit in long-range connections and an excess of short-range connections. *Neuropsychologia*. 2011.

Peer-Reviewed Articles (full list)

1. Herzog R, Lehue F, Legaz A, Azzi T, *et al.*, **Barttfeld P**, *et al.*, Ibáñez A. Biophysical brain twins map the phenotype-expotype architecture of aging and dementia. *Nature Biomedical Engineering*. 2026.
2. Sarli L, **Barttfeld P**, Justel N. Differential responses for emotional visual and auditory word recognition in first vs. second language. *International Journal of Bilingualism*. 2026.
3. Comay N, Solovey G*, **Barttfeld P***. Confidence phenotypes: A unified computational account of value and decision certainty in reinforcement learning. *eLife*. 2026.
4. Sarton B, Mattia GM, Cervoni E, Decourt J, Peran P, Riu B, Bounes F, Naboulsi E, **Barttfeld P**, Olivot J, Silva S, Cussat-Blanc S. Explainable Machine Learning for Coma Outcome Prediction Based on Structural and Functional Brain MRI. *Critical Care Medicine*. 2026.
5. Della Bella G, Vélez Picatto A, Galván Rial S, Cukier S, Foa Torres G, Catanzariti M, Mateos D, Lamberti P, Cardaña E*, **Barttfeld P***. The Neurophenomenology of a Self-Induced Transcendental Visionary State: A Case Study. *NeuroImage*. 2026.
6. Legaz A, Moguilner S, **Barttfeld P***, *et al.*, Hernández H, Ibañez A. The exposome of brain aging across 34 countries. *Nature Medicine*. 2026.
7. Della Bella G, Sarton B, Mattia G, Peran P, Lamberti W, **Barttfeld P***, Silva S*. Brain dynamic organization at the acute stage of severe brain injury. *NeuroImage*. 2025.
8. Comay N, Solovey G*, **Barttfeld P***. Decisions are based on less information than metacognitive judgments in multi-alternative contexts. *Journal of Experimental Psychology: LMC*. 2025.
9. Comay N, Solovey G, **Barttfeld P**. Iconic vs. working memory metacognition to evaluate the richness of perception: a registered report. *Royal Society Open Science*. 2025.
10. **Barttfeld P**, Comay N, Embon I, Solovey G. Pre- and post-decision signals of certainty in changing minds. *Trends in Neurosciences*. 2025.
11. Aguirre J, **Barttfeld P**, Martín-Arévalo E, Lupiáñez J, Luna F. Testing thought-probe frequency for measuring mind-wandering along with vigilance and cognitive control loss. *Behavioral Research Methods*. 2025.
12. Hazelton J, Migeot J, Gonzalez-Gomez R, *et al.*, **Barttfeld P**, Ibanez A. Cardiovascular risk factors and the allostatic interoceptive network in dementia. *Cardiovascular Research*. 2025.
13. Della Bella G*, Zang D*, Gui P*, Mateos D, Sitt J, Bekinschtein T, *et al.*, **Barttfeld P**. Detection of EEG dynamic complex patterns in disorders of consciousness. *Communications Biology*. 2025.
14. Cardaña E, Berkovich-Ohana A, Valli K, **Barttfeld P**, *et al.* Bringing Order to Disarray: A Consensus Taxonomy of Altered States of Consciousness. *Psychology of Consciousness*. 2025.

15. Gonzalez-Gomez R, Cruzat J, Hernández H, *et al.*, **Barttfeld P**, Ibañez A. Qualitative and quantitative educational disparities and brain signatures in aging and dementia. *eClinicalMedicine*. 2025.
16. Hazelton J*, Della Bella G*, **Barttfeld P**, *et al.* Altered spatiotemporal brain dynamics of interoception in behavioural-variant frontotemporal dementia. *eBioMedicine*. 2025.
17. Zapata J, Comay N, **Barttfeld P**, Solovey G, Saal A, Ahumada J. Metacognitive sensitivity on the Iowa Gambling Task reveals awareness as a necessary condition for advantageous performance. *Experimental Psychology*. 2025.
18. Legaz A, Altschuler F, Gonzalez-Gomez R, *et al.*, **Barttfeld P**, Ibañez A. Structural inequality linked to brain volume and network dynamics in aging and dementia across the Americas. *Nature Aging*. 2024.
19. Embon I, Gerbaudo MA, Ramos Usaj A, Iorio A, **Barttfeld P***, Solovey G*. Exploring the Relationship between Dysfunctional Personality Traits with Metacognition and Confidence. *Current Psychology*. 2024.
20. Ibanez A, Moguilner S, Baez S, *et al.*, **Barttfeld P**, Miller B. Brain clocks capture diversity and disparities in aging and dementia across geographically diverse populations. *Nature Medicine*. 2024.
21. Teitelbaum Dorfman F, Kogan B, **Barttfeld P**, García A. Bilinguals on the footbridge: The role of foreign-language proficiency in moral decision making. *Bilingualism: Language and Cognition*. 2024.
22. Legaz A, Prado P, Moguilner S, *et al.*, **Barttfeld P**, Ibañez A. Social and non-social working memory in neurodegeneration. *Neurobiology of Disease*. 2023.
23. Luna F, Aguirre J, Martín-Arévalo E, Lupiáñez J, **Barttfeld P**. Different oscillatory rhythms anticipate failures in executive and arousal vigilance. *Frontiers in Cognition*. 2023.
24. Embon I, Cukier S, Iorio A, **Barttfeld P***, Solovey G*. Is visual metacognition associated with ASD traits? *Consciousness and Cognition*. 2023.
25. Luna F, Aguirre J, Ibañez A, Martín-Arévalo E, Lupiáñez J, **Barttfeld P***. Event-related potentials associated with attentional networks evidence changes in executive and arousal vigilance. *Psychophysiology*. 2023.
26. Comay N, Della Bella G, Lamberti W, Sigman M, Solovey G*, **Barttfeld P***. The presence of irrelevant alternatives paradoxically increases confidence in perceptual decisions. *Cognition*. 2023.
27. Luna F, **Barttfeld P**, Martín-Arévalo E, Lupiáñez J. Cognitive load mitigates the executive but not the arousal vigilance decrement. *Consciousness and Cognition*. 2021.
28. Alcaide S, Sitt J, Horikawa T, Romano A, Maldonado A, Ibañez A, Sigman M, Kamitani Y, **Barttfeld P**. fMRI lag structure during waking up from early sleep stages. *Cortex*. 2021.
29. Luna F, **Barttfeld P**, Martín-Arévalo E, Lupiáñez J. The ANTI-Vea task: analyzing the executive and arousal vigilance decrements while measuring the three attentional networks. *Psicológica*. 2021.
30. Luna F, Román-Caballero R, Lupiáñez J, **Barttfeld P**, Martín-Arévalo E. Online High-Definition tDCS over the right fronto-parietal network improves phasic alertness and reduces the executive vigilance decrement. *Neuropsychologia*. 2020.
31. Demertzi A, Tagliazucchi E, Dehaene S, Deco G, **Barttfeld P**, Raimondo F, Rohaut B, Schiff N, Owen A, Laureys S, Naccache L, Sitt J. Human consciousness is supported by dynamic complex patterns of brain signal coordination. *Science Advances*. 2019.
32. Uhrig L, Sitt J, Jacob A, Tasserie J, **Barttfeld P**, Dupont M, Dehaene S, Jarraya B. Resting-state dynamics as a cortical signature of anesthesia in monkeys. *Anesthesiology*. 2018.

33. **Barttfeld P***, Abboud S*, Lagercrantz H, Adén U, Padilla N, Edwards AD, Cohen L, Sigman M, Dehaene S, Dehaene-Lambertz G. A lateral-to-mesial organization of human ventral visual cortex at birth. *Brain Structure and Function*. 2018.
34. Ais J, Zylberberg A, **Barttfeld P**, Sigman M. Individual differences in the shape, bias and precision of confidence distributions across perceptual decisions. *Cognition*. 2016.
35. **Barttfeld P**, Bekinschtein TA, Salles A, Stamatakis EA, Adapa R, Menon DK, Sigman M. Factoring the brain signatures of anesthesia concentration and level of arousal across individuals. *NeuroImage: Clinical*. 2015.
36. **Barttfeld P***, Uhrig L*, Sitt J*, Sigman M, Jarraya B, Dehaene S. Signature of consciousness in the dynamics of resting-state brain activity. *Proceedings of the National Academy of Sciences*. 2015.
37. Sedeño L, Couto B, Melloni M, *et al.*, **Barttfeld P**, Ibañez A. How do you feel when you can't feel your body? Interoception, functional connectivity and emotional processing in Depersonalization-Derealization Disorder. *PLoS One*. 2014.
38. **Barttfeld P***, Petroni A*, Baez S, *et al.* Functional Connectivity and Temporal Variability of Brain Connections in Adults with ADHD or Bipolar Disorder: A Default Mode Network Analysis. *Neuropsychobiology*. 2014.
39. Couto B, Salles A, Sedeño L, *et al.*, **Barttfeld P**, Ibañez A. The man who feels two hearts: The different pathways of interoception. *Social Cognitive and Affective Neuroscience*. 2013.
40. **Barttfeld P**, Amoroso L, Ais J, Cukier S, Bavassi L, Tomio A, Manes F, Ibañez A. Organization of brain networks governed by long-range connections index autistic traits in the general population. *Journal of Neurodevelopmental Disorders*. 2013.
41. **Barttfeld P**, Wicker B, McAleer P, Belin P, Cojan Y, Graziano M, Leiguarda R, Sigman M. Distinct patterns of functional brain connectivity correlate with objective performance and subjective beliefs. *Proceedings of the National Academy of Sciences*. 2013.
42. Gallos L, **Barttfeld P**, Havlin S, Sigman M, Makse H. Collective behavior and critical fluctuations in the spatial spreading of obesity, diabetes and cancer. *Scientific Reports*. 2012.
43. Zylberberg A, **Barttfeld P**, Sigman M. The construction of confidence in a perceptual decision. *Frontiers in Integrative Neuroscience*. 2012.
44. **Barttfeld P**, Wicker B, Cukier S, Navarta S, Leiguarda R, Sigman M. State-dependent changes of connectivity patterns and functional brain network topology in Autism Spectrum Disorder. *Neuropsychologia*. 2012.
45. **Barttfeld P**, Wicker B, Cukier S, Navarta S, Lew S, Sigman M. A big world network in ASD: Dynamical connectivity analysis reflects a deficit in long-range connections and an excess of short-range connections. *Neuropsychologia*. 2011.
46. Yorio A, Tabullo A, Wainseboim A, **Barttfeld P**, Segura E. Event-related potential correlates of perceptual and functional categories. *Neuroscience Letters*. 2008.
47. Abril A, **Barttfeld P**, Bucher E. The effect of fire and overgrazing disturbances on soil carbon balance in the Dry Chaco forest. *Forest Ecology and Management*. 2005.

Preprints

1. Galván Rial DS, Della Bella GA, Naci L, Delli Pizzi S, Sensi S, Osán T, Aguilera D, Timmermann C, Carhart-Harris R, Stamatakis EA, **Barttfeld P**. A Shared Entropic Axis Spans States of Consciousness Across Pharmacological and Clinical Conditions. *BiorXiv*. 2026.

Book Chapters

1. Fogliatti E, Galván Rial S, Della Bella G, **Barttfeld P**. Del ritual al protocolo: el conflicto en torno a los psicodélicos. In: A. Pamela Paz García (Ed.), *Drogas y bien público*. Institute of Psychological Research, Córdoba. 2025. (in Spanish). ISBN 978-987-47803-5-5.

FELLOWSHIPS AND AWARDS

- **INSERM Postdoctoral Fellowship** — INSERM, France 2013–2015
- **Raíces Return Fellowship** — MinCyT, Argentina 2015
- **James S. McDonnell Foundation Fellowship** — 2nd Latin American School of Cognitive and Neural Sciences 2012
- **CONICET Travel Fellowship** — University of Cambridge, UK 2012
- **Raúl Carrea Graduate Fellowship** 2010
- **CNRS Travel Fellowship** (ECOS project) — France 2009
- **Santa Fe Institute Summer School** — competitive international selection, Qingdao, China 2004

Training fellowships: doctoral, University of Buenos Aires (2004–2010); postdoctoral, CONICET (2011–2013).

GRANTS AND FUNDING

- **BeTA** — Erasmus+ Design Measures, European Commission 2026–2027
- **Royal Society International Exchanges Grant** 2026
PI: E.A. Stamatakis, University of Cambridge, UK
- **PICT 2022 (PI)** — ANPCyT, Argentina 2024–2026
- **PICT 2022 (PI)** — ANPCyT, Argentina 2024–2026
- **STIC-AmSud Programme (PI)** — ANID, Chile 2023–2024
- **Center for Open Science / Templeton Foundation (PI)** — Consciousness research 2023
- **PICT 2021 Argentina Innovadora 2020 (PI)** — ANPCyT 2021–2024
- **PICT 2018 (Co-investigator)** — ANPCyT 2018–2021
- **Emerging Research Groups Grant (PI)** — MinCyT Córdoba 2018
- **CONICET Short Research Stay Grant for Early-Career Researchers** 2018
- **Primar Grant (Co-PI)** — UNC 2018
- **PICT Young Investigators (PI)** — ANPCyT 2015–2018

RESEARCH SUPERVISION

Postdoctoral Researchers

- Gabriel Della Bella (Physics, UNC) 2026–2029
- Santiago Redondo (Psychology, UNC) 2024–2026

Doctoral Students

- Sebastián Galván Rial (Neuroscience, UNC) expected 2029
- Daniel Aguilera (Psychology, UNC) expected 2028
- Julieta Aguirre (Psychology, UNC) expected 2026
- Iair Embón (Neuroscience, UNC) completed 2026
- Nicolás Comay (Neuroscience, UNC) completed 2026
- Julieta Zapata (Psychology, UNC) completed 2025
- Gabriel Della Bella (Physics, UNC) completed 2025
- Renato Paredes (Neuroscience, UNC) completed 2025
- Matías Palmucci (Neuroscience, UNC) completed 2025
- Leticia Sarli (Neuroscience, UNC) completed 2025
- Federico Cavanna (Neuroscience, UNC) completed 2025
- Fernando Luna (Psychology, UNC) completed 2019

Master's Students

- Julieta Irazuzta, Vera Candusso, Sebastián Galván Rial (Psychology, UNC) 2025

• Federico Teitelbaum Dorfman (Psychology, UNC)	2022
• Rocío Rodríguez (Psychology, UNC)	2021
• Nicolás Comay (Psychology, UNC)	2020
• Julieta Aguirre (Psychology, UNC)	2019
• Santiago Alcaide, Álvaro Romano (Biomedical Engineering, UNC)	2018
• Marcos Grillo (Psychology, Favaloro University)	2018
• Joaquín Ais — co-supervised (Biology, UBA)	2016

More than 60 students completed their undergraduate final research project in the laboratory, supervised by me or by members of the group.

RESEARCH STAYS

• Institut du Cerveau et de la Moelle Épineière , Paris, France	2019 · 3 months
• Addenbrooke's Hospital , University of Cambridge, UK	2012 · 2 months
• Institut de Neurosciences Cognitives de la Méditerranée , Marseille, France	2009 · 2 months

EDITORIAL AND PEER-REVIEW ACTIVITY

Editorial Roles

• Associate Editor , <i>Imaging Neuroscience</i>	2023–present
• Associate Editor , <i>NeuroImage: Reports</i>	2024–present

Ad hoc Reviewer

Nature Communications, Communications Psychology, Communications Biology, Cerebral Cortex, Communications Medicine, Journal of Neuroscience, NeuroImage, Human Brain Mapping, Frontiers in Human Neuroscience, Frontiers in Aging Neuroscience, Brain and Cognition, CABN, Translational Psychiatry, Journal of Neurophysiology, Journal of Psychiatric Research, Developmental Psychology, among others.

Invited Expert Review

• Book proposal reviewer, <i>Springer Nature</i> (Neuroscience)	2026
• Metascience Novelty Indicators Challenge, University of Sussex	2026

GRANT AND THESIS EVALUATION

Grant Review Panels

• CONICET Researcher Career admissions , Argentina	2026
• Member, Psychology Fellowship Committee , CONICET, Argentina	2026
• María Viñas Fund , ANII, Uruguay (2 proposals)	2026
• Research group evaluation , CSIC, Universidad de la República, Uruguay	2026
• NWO Talent Programme Vici , Dutch Research Council	2025
• Fédération pour la Recherche sur le Cerveau , France	2024
• CONICET , Argentina	2017, 2019–2024

Doctoral Theses Examined (external examiner)

• Juan Cruz Berón — University of Buenos Aires	2026
• Gustavo Juantorena — University of Buenos Aires	2026
• Nerea Herrero — National University of the Center of Buenos Aires Province	2026
• Paul Jacquart — Université de Toulouse, France (thesis review committee)	2026
• Soledad Picco — Biology, University of Buenos Aires	2026
• María Laura Noguera — Medicine, Catholic University of Cuyo	2025
• Raúl González Gómez — Neuroscience, Adolfo Ibáñez University, Chile	2024
• Agustina Legaz — Psychology, UNC	2024
• Fabricio Baglivo — Engineering, UBA	2024
• Marcos Pietto — Psychology, UNC	2020

- Noelia Calvo — Psychology, UNC 2019
- Débora Mola — Psychology, UNC 2019
- Soledad Picco — Biology, UBA 2018
- Santiago Redondo — Psychology, UNC 2018
- Lucas Sedeño — Medicine, Favaloro University 2018
- María González Gadea — Neuroscience, UNC 2015

I have also examined more than 20 Master's theses at UNC, UBA and Favaloro University.

TEACHING

Introduction to Research Practice in Cognitive Science 2019–2024

School of Psychology, UNC, Argentina

Undergraduate · experimental methods, task design, behavioural data analysis.

Practical Introduction to Cognitive Science (Graduate) 2023

School of Psychology, UNC, Argentina

Graduate · computational modelling and analysis of brain time series.

Biological Systems (Teaching Assistant, Electronic Engineering programme) 2005–2018

School of Engineering, University of Buenos Aires

SELECTED PRESENTATIONS

- **Invited talk**, School of Neuroscience and Behavioural Disorders, Montevideo, Uruguay 2024
- **Invited talk**, Workshop on Sleep, Dreaming and Consciousness, Buenos Aires, Argentina 2023
- **Invited talk**, Institute of Applied Mathematics (IMAL), Santa Fe, Argentina 2019
- **Invited talk**, *Mind, Society and Behavior* Conference, CONICET, Buenos Aires 2018
- **Invited talk**, 2nd Meeting of the Latin American Brain Mapping Network, Buenos Aires 2017
- **Invited talk**, INECO International Symposium “New Trends in Neurodevelopment and Cognition”, Buenos Aires 2015
- **Invited talk**, “Brain Dynamics in the Primate Brain” Conference, NeuroSpin, Saclay, France 2015
- **Invited talk**, First Theory–Cognition Workshop, European Institute for Theoretical Neuroscience, Paris 2014

SERVICE AND ADMINISTRATION

- **Scientific Committee Member**, 29th Annual Meeting of the Association for the Scientific Study of Consciousness (ASSC), Chile 2025–2026
- **Board Member**, Institute of Psychological Research – CONICET 2025
- **Ethics Committee Member**, Institute of Psychological Research – CONICET 2021–2024